Week 02 Reading: Climate as a dynamical system

Readings on dynamical systems, autocorrelation, and spectra

2026-08-31

Submit your answers on Canvas by Sunday 8/30 at 11:59 pm, before Monday's class.

Assigned

1. Section 3.1 of Kantz (2004), stopping at the end of Example 3.3, which defines a dynamical system as a state and a rule that advances it. This reading is available through the Rice library; search the ISBN 978-1-107-13655-7 at library.rice.edu.
2. Sections 1, 2.1, and 2.2.2 of Ghil et al. (2011), which give a working definition of an extreme event. The reading explains why a system cannot be reconstructed from one short record and decomposes a spectrum into a continuous background and discrete lines. This reading is open access.
3. Sections 2.1 through 2.3 of Kantz (2004), to skim, which cover stationarity, the autocorrelation function, and the periodogram. We derive this material at the board on Wednesday, so read past anything you cannot follow.

Questions

Write a few sentences on each.

1. In your own words, what is a dynamical system? Give an example from your own field, name its state and its rule, and say whether you could actually write the rule down.
2. Kantz (2004) says that autocorrelations cannot distinguish random from deterministic chaotic signals. Why does that matter for someone estimating flood risk from a gauge record?
3. Section 2.1 of Ghil et al. (2011) explains why reconstructing a system's behavior works on numerically generated data and fails on real geophysical records. What is the difference, and why does it matter for a 100-year record?
4. A spectrum shows a peak sitting on a wiggly background. Name two kinds of process that could have produced this spectrum.

Bibliography

Ghil, M., Yiou, P., Hallegatte, S., Malamud, B. D., Naveau, P., Soloviev, A., et al. (2011). Extreme Events: Dynamics, Statistics and Prediction. *Nonlinear Processes in Geophysics*, *18*(3), 295–350. https://doi.org/10/fvzxvv

Kantz, H. (2004). *Nonlinear Time Series Analysis* (Second edition.). Cambridge: University Press.